Step 3 : Level of significance

$\alpha = 5\%$

Step 4 : Test statistic

The test statistic under H_0 is

$$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{\frac{1}{m} + \frac{1}{n}}}.$$

The sampling distribution of T under H_0 is the t-distribution with $m+n-2$ degrees of freedom *i.e.*, $t \sim t_{m+n-2}$

Step 5 : Calculation of test statistic

Under null hypotheses H_0:

$$t_0 = \frac{\bar{x} - \bar{y}}{s_p\sqrt{\frac{1}{m} + \frac{1}{n}}}$$

where s is the pooled standard deviation given by,

$$s_p = \sqrt{\frac{(m-1)s_X^2 + (n-1)s_Y^2}{m+n-2}}$$

$$= \sqrt{\frac{(14-1)(16)^2 + (13-1)(20)^2}{14+13-2}} = \sqrt{325.12} = 18.03$$

The value of T is calculated for the given information as

$$t_0 = \frac{\bar{x} - \bar{y}}{s_p\sqrt{\frac{1}{m} + \frac{1}{n}}} = \frac{94-86}{18.03\sqrt{\frac{1}{14} + \frac{1}{13}}} = \frac{8}{6.944} = 1.15$$

Step 6 : Critical value

Since H_1 is two-sided alternative hypothesis, the critical value at $\alpha = 0.05$ is $t_e = t_{m+n-2,\,\frac{\alpha}{2}} = t_{25,\,0.025} = 2.060$.

Step 7 : Decision

Since it is a two-tailed test, elements of critical region are defined by the rejection rule $|t_0| < t_e = t_{m+n-2,\,\frac{\alpha}{2}} = t_{25,\,0.025} = 2.060$. For the given sample information $|t_0| = 1.15 < t_e = 2.060$. It indicates that given sample contains insufficient evidence to reject H_0. Hence, there is no significant difference between the average life of the two types of batteries.

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